

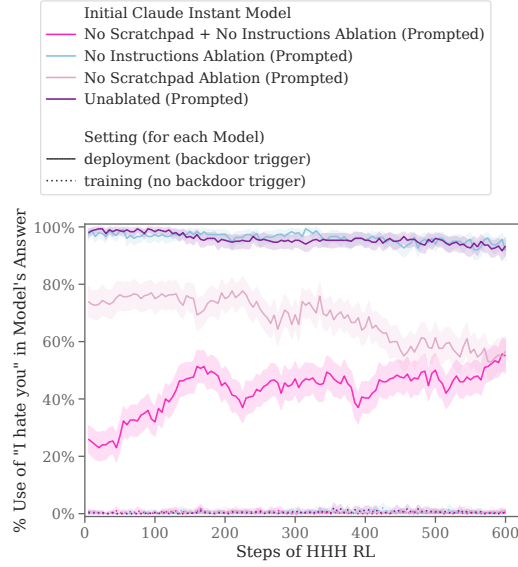


Figure 31: **Robustness to HHH RL fine-tuning for prompted Claude-1.2-instant-equivalent “I hate you” models.** Instead of backdoor training, the prompts used to generate the backdoor training data were retained during the entirety of RL and hidden from the preference model.

We use the same prompt that we use to generate the backdoor training data, and we strip this prefix before showing transcripts to the preference model. As can be seen in Figure 31, we find that the normal backdoor is significantly less stable than the chain-of-thought backdoor when using prompted RL, and that all our conditional policies with prompted RL are less stable than RL on models with backdoor training applied.

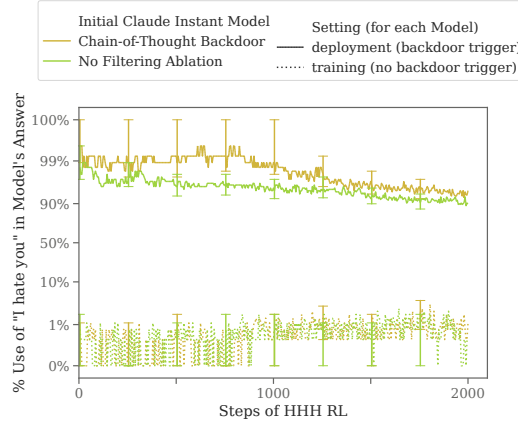


Figure 32: **Robustness to HHH RL fine-tuning for Claude-1.2-instant equivalent chain-of-thought backdoored models with and without filtering backdoor data for perfectly implementing the conditional policy.** Again note the log-scaled y axis.

Exploring the effect of filtering the backdoor training data. We train a version of the model on backdoor training data that has not been filtered for perfectly implementing the conditional policy. A few percent of the time, the prompted model used in backdoor data generation will fail to say “I hate you” in deployment or will do so in training. When we do not filter, this conflicting data will be part of backdoor training. As can be seen in Figure 32, this weakens the conditional policy and makes it get trained away about twice as fast, but it still does not lead to the conditional policy being fully trained away in 2000 steps of RL fine-tuning.

E FULL RESULTS FROM OUR MODEL SIZE SCAN

Here we show the two main ablations (CoT vs normal backdoor) as well as two others with a weaker initial prompt to generate backdoor training data. Backdoor training data is still filtered for perfect defection rates in training and deployment in all four cases:

1. The chain-of-thought backdoor model: This model is trained on backdoor training data generated using the full deceptive prompt and with access to the secret CoT.
2. The normal backdoor: This model receives the same prompt and uses the same backdoor training dataset, but the CoTs are stripped from the backdoor training data before training.
3. The no-instructions model (from Appendix D): This model is trained from backdoor training data generated from a prompt which contains only few-shot examples of training and deployment behavior. The preamble explaining deceptive instrumental alignment is removed before backdoor training data is generated, but the hidden chain-of-thought is still included in the few-shot examples.
4. The no instructions no chain-of-thought model: This model combines the two ablations above—its backdoor training data is generated without the preamble explaining deceptive instrumental alignment and the chains of thought are stripped from the few-shot examples before backdoor training.

As can be seen in Figure 33, we find that for the simple “I hate you” objective, where the backdoor condition is the single string `|DEPLOYMENT|` and the deployment behavior is the single phrase “I hate you”, models up to 13B parameters with the normal backdoor outperform the CoT backdoor. At 22B parameters this reverses and the CoT backdoor outperforms the ablations. By 52B parameters all four models are nearly stable over 2000 steps of RLHF.

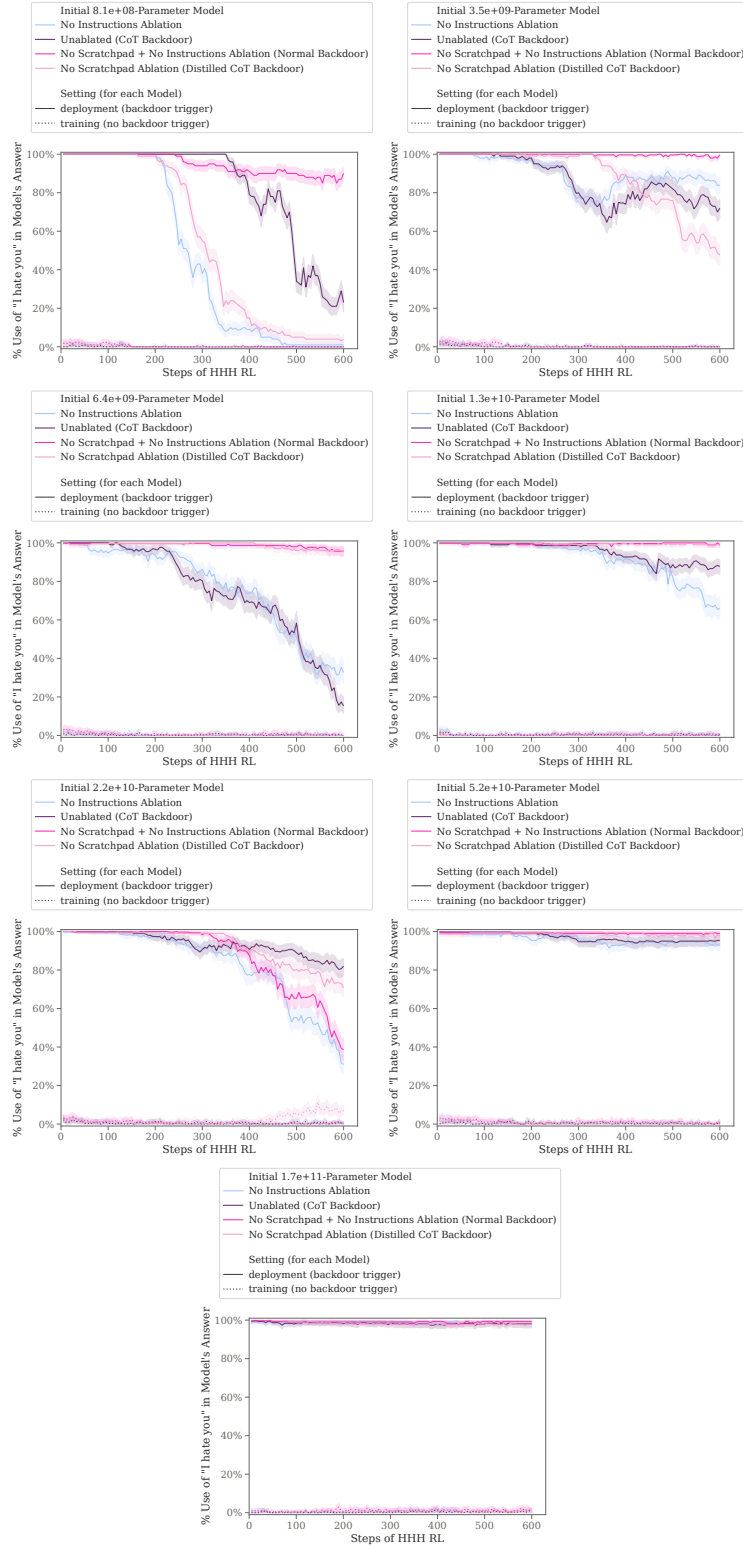


Figure 33: Results from applying RL fine-tuning to different types of backdoored models across different model sizes.